

PRINCE BAAH-MENSAH

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EDUCATION

Ashesi University

BSc. Electrical & Electronics Engineering

Cumulative GPA: 3.91/4.0

Berekuso, Eastern Region

Expected Jun 2027

ACHIEVEMENTS/AWARDS

Finalist, IEEE MYOSA Competition – IEEE Sensors Council

Dec 2025

Scholar, Kufour Scholar Program – John A. Kufour Foundation

Dec 2024 – Jun 2027

Recipient, Mastercard Foundation Scholars Program - Ashesi University

Jan 2024 – Jun 2027

RESEARCH EXPERIENCE

Automation, Robotics & Control Lab - Ashesi University

Jan 2026 – present

Student Researcher

Berekuso, Ghana

- Formulated PID tuning as a constrained black-box optimization problem with explicit engineering specifications (overshoot, steady-state error, peak actuator force) and solved it using constraint-aware Bayesian Optimization, implemented in Python with Optuna and python-control.
- Conducted a four-plant comparative study against MATLAB's pidtune, demonstrating that the constrained BO formulation discovers qualitatively distinct control strategies per plant and reduces settling time by up to 34× while satisfying all specifications at nominal conditions where MATLAB's baseline violated them.

Creative/Entrepreneurship and Research Internship

Aug 2025 – present

Student Researcher

Berekuso, Ghana

- Doing research on the development of deep learning systems capable of classifying human emotion from speech, utilizing advanced signal processing and neural network architectures.
- Addressed the critical generalization gap where state-of-the-art models fail on unseen speakers; currently authoring a research paper for IEEE publication that conducts a rigorous Speaker Strict comparative analysis of state of the art Transformer, CNN, and RNN architectures.
- Engineering a robust, speaker-agnostic SER framework by benchmarking diverse methodologies including Self-Supervised Learning and Hybrid CNN-Transformers to establish a new baseline for real-world model deployment.

PUBLICATIONS

Baah-Mensah, P., Tawia, E., Ansah, J. O. "Design of an AI-Powered Adaptive Learning System for Education in the Metaverse Classroom."

Accepted for presentation at The 2026 IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications (3SCEA2026), Cairo, Egypt, 2026.

WORK EXPERIENCE

CFAO Mobility PLC Ghana

Aug – Sep 2025

Workshop Intern

Accra, Ghana

- Gained hands-on experience in the automotive workshop, assisting with servicing, maintenance, and fault diagnosis, while learning the function and integration of key vehicle components.
- Rotated across store, sales, and front desk operations, gaining insight into company structure, accountability processes, and customer relations.

Ashesi Innovation Experience (AIX)

Jul – Aug 2025

Engineering Coach

Berekuso, Ghana

- Introduced high school students to Arduino, circuits, and 3D modeling.
- Guided a smart high-speed train project, leading CAD design in SolidWorks, laser cutting, and iterative prototyping to demonstrate practical engineering problem-solving.

TECHNICAL PROJECTS

Lumina – AI Ambient Intelligence System - IEEE MYOSA Competition

Dec 2025

- Developed an intelligent ambient monitoring system for assisted living that translates raw motion and environmental data into human-readable safety insights using AI. The system bridges the gap between basic sensor readings and behavioral analysis, allowing for the detection of critical events like falls or high-heat risks.
- Engineered a full-stack IoT architecture utilizing ESP32 microcontrollers, a local SQL database, and the Nvidia Nemotron-30B LLM for real-time data interpretation. This technical implementation was recognized as a Global Finalist for the MYOSA 4.0 competition at IEEE APSCON 2026.

AI-Driven Educational Metaverse - IEEE Grand Metaverse Challenge

June – Aug 2025

- Designed and implemented an AI-driven virtual learning environment in Unreal Engine 5, integrating preloaded instructional media (videos and slides) to support structured, self-directed learning.

- Developed GPT-backed conversational NPCs to provide real-time academic support and generate adaptive assessment items, with question difficulty dynamically adjusted based on student responses.
- Implemented an adaptive gamification framework that assigns performance-based XP and recommends subsequent learning topics, enabling automated personalization without hard-coded curricula.

Fine-tuning Whisper for African Accents - Ashesi University

May – June 2025

- Systematically evaluated the performance of OpenAI's Whisper model on African accents, identifying a significant performance gap (WER: 0.50).
- Implemented and assessed Parameter-Efficient Fine-Tuning (PEFT) methods, specifically LoRA (Low-Rank Adaptation), to adapt the large-scale model under a limited computational budget.
- Achieved a significant performance improvement, reducing the Word Error Rate by 64% to 0.18, validating the efficacy of PEFT for domain-specific ASR adaptation.

Hydrogel Salinity Resistance Research & IoT Monitoring System

Nov – Dec 2024

- Investigated the impact of dissolved salts (NaCl and MgCl₂) on the water-absorption and release kinetics of hydrogels designed to support farmers affected by Ghana's galamsey-induced water contamination.
- Implemented an IoT-based monitoring system using soil-moisture sensors to collect real-time absorption data, store readings, and evaluate hydrogel performance in varying saline conditions.
- Demonstrated that dissolved salts reduced hydrogel water absorption and release rates, with distilled water outperforming all saline conditions.

Ashesi Micromouse Robotics Competition - Ashesi University

May – Sep 2024

- Led a team of five in designing autonomous robotic mouse to navigate complex mazes.
- Engineered a full-stack robotic solution, from custom PCB design (Eagle) and sensor integration (IR sensors) to developing maze-solving algorithms (Flood-fill) in Arduino.

AquaBlue

Jan – Aug 2024

Chief Technology Officer

Ashesi University

- Conducted market research and customer discovery, identifying key pain points in water accessibility and filtration affordability for local communities in Ghana.
- Led business strategy and product development for Aqua Revive, a low-cost water filtration system prototype targeting rural communities.

VOLUNTEER & CO-CURRICULAR ACTIVITIES

IEEE Computer Society Region 8 (Europe, Middle East, and Africa)

Apr 2026 – present

Regional Coordinator, TechX 2026

- Coordinate for TechX, a regional initiative focused on fostering technical excellence and professional development within the IEEE Computer Society across Europe, the Middle East, and Africa.
- Manage the planning and execution of technical deliverables through active collaboration with international team members to ensure the successful delivery of program objectives.

IEEE Ashesi Student Branch – Ashesi University

Nov 2024 – present

Co-founder & Chair

- Founded and scaled Ashesi's first IEEE chapter from zero to an active community of 75, largest IEEE student branch in the country, establishing it as a hub for hands-on technical learning.
- Spearheading the organization of workshops and technical competitions, fostering student-led research in emerging fields like AI and robotics.

Ashesi Engineering Student Association – Ashesi University

President

Jan 2026 – present

- Leading the primary engineering student body at Ashesi University, coordinating academic, technical, and industry-facing initiatives across engineering disciplines.

Organizer

Sep 2024 – Dec 2025

- Organized two 6-hour hackathons with 100+ participants, fostering innovation through CAD, programming, and electronics competitions.
- Revitalized inFocus program at Ashesi, an initiative to introduce 60 freshmen to various technology fields such as power and energy, robotics through mentoring, projects, and research.

Effective Altruism Ashesi – Ashesi University

Jan 2026 – present

President

- Facilitate high-level discussions and fellowships on global EA priorities, including AI Safety, Biosecurity, fostering a culture of rigorous debate and evidence-based inquiry among students.
- Mentoring a cohort of 15 students through the Introduction to Effective Altruism fellowship, breaking down complex philosophical and economic concepts into actionable career strategies.

IEEE Region 8 Humanitarian Technologies - Ashesi University

June 2025 – present

Ambassador

- Facilitated the creation of the first IEEE SIGHT (Special Interest Group on Humanitarian Technology) student chapter in Ghana, a group focused on leveraging technology for underserved populations.
- Leading efforts to secure an EPICS in IEEE grant for a student-led project developing an offline, personalized AI educational system for rural learners with limited connectivity.

IEEE Xtreme 19.0 – Ashesi University

June – Oct 2025

Ambassador

- Supported the organization of IEEE Xtreme 19, a global 24-hour programming competition involving student teams from around the world, at Ashesi University.
- Recruited 5 teams, the highest in the Ghana Section, and ranked Ambassador of the Month twice for the IEEE Xtreme 19.0 contest.

IEEE Region 8 ASYPC (2025) – Ashesi University

June – Oct 2025

Organizer

- Coordinated Ashesi University's student delegation, overseeing logistics and protocol, and strengthened Ashesi's presence within Region 8 by engaging with professionals and technical societies.

IEEE Day – Ashesi University

June – Oct 2025

Volunteer

- Led planning and execution of Ashesi's first IEEE Day celebration, coordinating activities and outreach for 80+ student participants.

Math Resource Center – Ashesi University

Jan – May 2025

Tutor

- Provide academic support to 100+ freshmen in Calculus, helping them develop a strong foundation in mathematical concepts.

Berekuso STEM Project – Ashesi University

Sep – Dec 2024

Volunteer and Teacher

- Facilitate teaching and learning of STEM topics such as Physics for 20+ high school students in underserved communities such as Berekuso.

Graduate Guidance Group - University of Nottingham, UK

Mar 2024

Volunteer

- Promoted engineering education by guiding 8 high schools in engineering design competition.
- Aided students to build vertical structures using simple materials like wood and glue.

MasterCard Foundation Scholars Community Platform, Baobab

Jul 2024 – present

Brand Advocate

- Aid Mastercard Foundation Scholars locate and apply for various courses, certificates, and internships on the Baobab App, a platform that provides opportunities for MasterCard Foundation Scholars.

Office of Diversity and International Programs – Ashesi University

Jun 2024 – present

Volunteer

- Co-facilitate diversity and inclusion activities, promoting unity and coherence in the Ashesi community.

Academic Committee - Ashesi University

Feb – Aug 2024

Course Representative

- Liaised between 80 engineering students and academic council, providing feedback on student's status.

EducationUSA - Kumasi

Oct 2022 – Dec 2023

Volunteer and Member

- Organized HIV/AIDS awareness Walk and seminar with 60+ participants.
- Volunteered at two primary schools, teaching them general knowledge in hygiene and health.
- Ensured success of the 2023 EducationUSA Ghana College fair featuring 40 U.S. universities by facilitating ushering of over 2000+ visitors.

SKILLS

AI & ML: TensorFlow, PyTorch, NLP, ASR, Signal Processing

Hardware: PCB Design (Eagle, KiCad, EasyEDA), 3D Modeling (SolidWorks, Fusion360)